

Deterministic loop-class mapping algorithm

Input: physical observable; outputs are uniquely determined by topology factors (n, g, s) .

Spinor-trace component $n \in \{0, 1, 2, 4\}$

Generation range $g \in \{0, 1/N_{\text{gen}}, 1/(2N_{\text{gen}})\}$

Sync coupling $s \in \{0, \varepsilon^2, \varepsilon^2 \text{ pure}\}$

Lemma 1
 $1 \pm \gamma/4$

Lemma 2
 $1 \pm \gamma^2/4$

Lemma 3
 $1 \pm \gamma^2$

Lemma 4
 $1/(1 \pm 2\gamma^2)$

Lemma 5
 $1 \pm \gamma/N$

Lemma 6
 $1 \pm \gamma/(2N)$

Lemma 7
 $1 \pm \gamma\varepsilon^2$

Lemma 8
 $1 \pm \gamma/2$

Pure-sync
 $1 \pm \varepsilon^2$

Sign + for CP-/T-even, – for CP-/T-odd (T-parity source axiom on the defect field)